

Sarthak Singh

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EXPERIENCE

Influenced (Full Stack Developer)

Next.js, TypeScript, Tailwind CSS, AWS Amplify, Dynamo DB, LLM

July 2024 – June 2025

Remote, USA

- Built serverless backend on AWS Lambda and API Gateway with 20+ endpoints handling product listings, interest workflows, RBAC.
- Containerized and deployed the full application on AWS using Docker, K8, and Amplify Gen 2 with CI/CD pipeline, integrating Cognito authentication, S3 image storage, and DynamoDB for a production-ready, scalable architecture.

IBM (Software Development Engineer II)

Python, AWS, GO, Java, Docker, Kubernetes, PostgreSQL, Nginx

July 2021 – Jan 2024

Kochi, India

- Led the migration of data analytics pipelines from ElasticSearch to OpenSearch, reducing operational costs by up to \$3M annually and improving search performance.
- Streamlined backend services for API management on IBM Cloud and AWS, optimizing traffic routing, security, and data processing, reducing inefficiencies and operational overhead.
- Spearheaded a security optimization project for managing sensitive data, resulting in a more cost-effective solution and earning the IBM Quarterly Cash Award for contributions.
- Enhanced the overall reliability and scalability of services, facilitating API integrations and improving customer-facing application performance.

Informatica (Software Engineer Intern)

Python, ETL, Oracle DB, Tableau, Java, AWS S3

Jan 2021 – July 2021

Bangalore, India

- Contributed to customer support by resolving 100+ tickets, offering technical guidance and troubleshooting for issues related to the Informatica ETL tool.
- Provided direct technical support for integrating the Informatica ETL tool with databases such as PostgreSQL, MySQL, and AWS S3.

PROJECT & RESEARCH

AI Powered Solutions

LLM, RAG, Gradio, Streamlit, Groq, MCP, HF Pipelines, LangChain

Oct 2025 – Dec 2025

[Link](#)

- RAG Voice Customer Support: An end-to-end voice-enabled customer support assistant combining Retrieval-Augmented Generation with real-time speech interaction.
- An AI-powered Indian Law Assistant leveraging RAG to ground responses in Indian legal statutes, utilizing MCP for seamless integration and processing of real-world legal data and case law analysis.
- GPT-OSS 120B CritiqueLoop: An automated critique → improve → repeat pipeline using Groq for ultra-fast inference.

Anomaly Detection in IoT Devices

[Link](#)

Aug 2024 – Dec 2024

- Explored unsupervised learning algorithms for anomaly detection in IoT devices, eliminating the need for labeled data while achieving strong detection performance.

Research Article (Journal of Environmental Management)

[Link](#)

July 2022 – Jan 2024

- Contributed to research on green procurement dynamics in the public sector, developing an assessment framework to help developing nations make data-driven decisions towards green environmental practices.

EDUCATION

University of Maryland

College Park, MD

Master in Engineering, Software Engineering, Cumulative GPA: 3.6/4.0

Jan 2024 – Dec 2025

Teaching Assistant | Introduction to Software Engineering ENPM611

Vellore Institute of Technology

Vellore, India

Bachelor of Technology in Computer Science and Engineering

June 2017 – May 2021

TECHNICAL SKILLS

Programming: Python, GO, FastAPI, Java, SQL (PostgreSQL, MySQL), NoSQL, Redis, Kafka, Django, GraphQL, Nginx
DevOps & Cloud: Docker, K8, Jenkins, CI/CD, Terraform, AWS(CloudWatch, Grafana, EC2, EKS, IAM, S3, Lambda)
Systems: Microservices, Distributed Systems, Scaling Systems, Object-Oriented Design, Unit Testing
Other: Git/GitHub, Windows, Linux, Data Structures & Algorithm, Communication, Innovation
AI: Cursor, NLP(transformer models), LLM(Claude,GPT), Deep Learning ML (Pytorch, SGD, scikit learn)